PH7002: Concepts in Statistical Mechanics
§6 Series Expansions

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§6.1 Aperitifs

§6.2 Perturbation theory

§6.3 Heuristic argument

§6.4 Probability theory

§6.5 Linked cluster theorem
Example 6.1: Gaussian distribution

§6.6 Wick's theorem for single variable

§6.7 Case of many random variables
Example 6.2: Joint Gaussian distribution

§6.8 Wick's theorem for many variables

§6.9 Relation to Statistical Mechanics

§6.10 Low temperature expansion

Example 6.3: Low temperature expansion of the Ising model

§6.11 High temperature expansion

Example 6.4: Spin-spin correlation function

Example 6.5: 2d spin 1/2 Ising model

§6.12 Self-duality in the two-dimensional Ising model

§6.13 Analysis of series expansion

§6.14 Extra notes

Reading

Yeomans chapter 6 and Sethna section 8.3.

Most statistical mechanical systems *cannot* be solved explicitly. Statistical mechanics does provide general relationships and organizing principles (temperature, entropy, free energy, thermodynamic relations) even when a solution is not available. But there are times when specific answers about specific models or experiments are needed.

There are two basic tools for extracting answers out of statistical mechanics for realistic systems. The first is **simulation**. Sometimes one simply mimics the microscopic theory. For example, a **molecular dynamics** simulation will move the atoms according to Newton's laws. If one is not interested in the detailed dynamical trajectories of the system, one can use **Monte Carlo simulation** methods to extract the equilibrium properties from a model. The theory underlying the Monte Carlo method is the mathematics of Markov chains, which arise in many other applications.

The second tool is to use **perturbation theory**. Starting from a solvable model, one can calculate the effects of small extra terms; for a complex system one can extrapolate from a limit (like zero or infinite temperature) where its properties are known. **Lattice models** such the Ising model can be studied by **series expansions**. In Physics, this involves computing **Feynman diagrams**. Such expansions start with certain exactly solvable limits, and typically represent perturbations around such limits by graphs on the lattice.

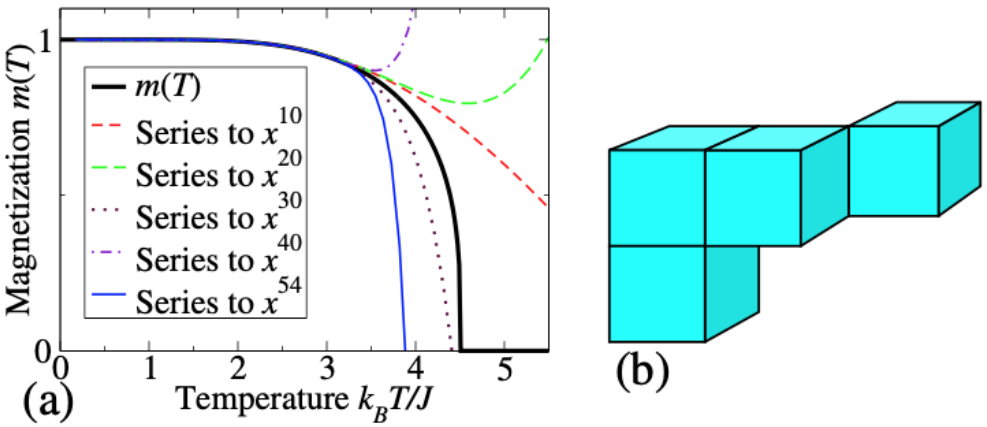
Important themes

1. Moment generating function and Cumulant generating function (Z vs $\ln Z$).
2. Linked cluster theorem (in a diagrammatic sense).
3. Low-temperature expansion for the Ising model.
4. High-temperature expansion for the Ising model.
5. Krammer-Wannier duality.

Perturbation theory

The Ising model is ferromagnetic below T_c and paramagnetic above T_c. Inside phases the properties do not shift in a singular way; one can smoothly extrapolate the behavior inside a liquid or magnetic phase under small changes in external conditions. **Perturbation theory works inside phases**. More precisely, inside a phase the properties are analytic (have convergent Taylor expansions) as functions of the external conditions. Lattice theories at high and low temperatures T have **perturbative expansions** in powers of $1/T$ and T, with Feynman diagrams involving all ways of drawing clusters of lattice points. Gases at high temperatures and low densities have **virial expansions**. Metals at low temperatures have Fermi liquid theory, where the electron-electron interactions are perturbatively incorporated by dressing the electrons into quasiparticles. Properties of systems near continuous phase transitions can be explored by perturbing in the dimension of space, giving the ϵ-expansion.

Fig. 8.7 Perturbation theory. (a) Low-temperature expansions for the cubic Ising model magnetization (Fig. 8.2) with successively larger numbers of terms. (b) The high- and low-temperature expansions for the Ising and other lattice models are sums over (Feynman diagram) clusters. At low T, Ising configurations are small clusters of up-spins in a background of down-spins (or vice versa). This cluster of four sites on the cubic lattice contributes to the term of order x^{20} in eqn 8.19, because flipping the cluster breaks 20 bonds.



For example, the low-temperature expansion of the magnetization per spin of the cubic-lattice three- dimensional Ising model starts out

$$m = 1 - 2x^6 - 12x^{10} + 14x^{12} - 90x^{14} + 192x^{16} - 792x^{18} + 2148x^{20} + \ldots,$$

where $x = e^{-2J/k_B T}$ is the probability to break a bond (parallel energy $-J$ to antiparallel energy $+J$).

This **series** was generated by carefully considering the probabilities of low-energy spin configurations, formed by flipping combinations of clusters of spins. This expansion smoothly predicts the magnetization at low temperatures using the properties at zero temperature. We see that the magnetization is well described by our series for $k_B T \lesssim 3J$.

No finite power series, however, can extrapolate past the temperature T_c at which the magnetization goes to zero. This is easiest to see in the opposite direction; $m(T) \equiv 0$ above T_c, so any extrapolation below T_c must continue to have zero magnetization. Much of the glory in perturbation theory involves summing infinite families of terms to extrapolate through critical points. **Phase boundaries** occur at parameter values where the properties are not smooth–where the continuation of the properties on one side does not predict the behavior on the other.

Consider a one-dimensional chain with free ends. The Hamiltonian in zero field is

with ground state energy $E_0 = -(N-1)J$. Suppose that the system is at a very low temperature and consider the class of excitations defined by $s_i = 1$, $i \leq \ell$ and $s_i = -1$, $i > \ell$:

There are $N-1$ such states, all with the same energy $E = E_0 + 2J$.

At temperature T the free energy change due to these excitations is

which is less than zero for all $T > 0$ in the limit $N \to \infty$. These excitations disorder the system; the expectation value of the magnetization is zero. Therefore, there cannot be a phase transition to a ferromagnetic state in the one-dimensional Ising model with nearest-neighbor (or with any finite-range) interactions.

A similar argument can be used to give a crude estimate of the transition temperature for the two-dimensional Ising model. Consider an $N \times N$ square lattice with free surfaces with the set of excitations of the type shown below.

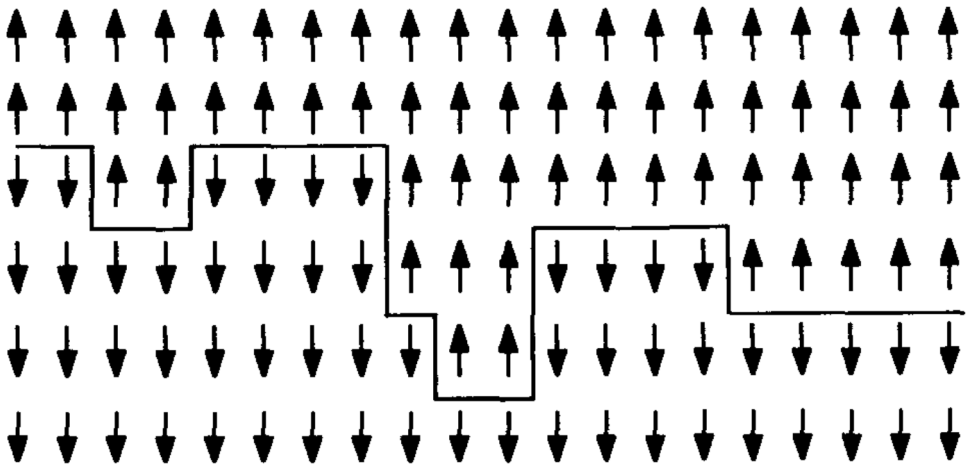


Figure: Domain wall in a two-dimensional Ising model.

These excitations which divide the lattice into two large domains separated by a wall that extends from one side to the other and has no loops. The energy of the domain wall is

where L is the number of segments in the wall. If we start the wall from the left there are at least two, sometimes three choices for the direction of the next step if we neglect the possibility of reaching the upper and lower boundaries. The entropy associated with non-looping chains of length L is then at least $k_B \log 2^L$. There are N possible starting points for the chain. If we assume two choices per site, one of which takes us to the right, the average length of the chain will be $2N$ with a standard deviation $\propto N^{1/2}$, which is small compared to N if N is large enough.

The free energy associated with dividing the lattice into two domains is thus approximately

The system is therefore stable against domain formation if

This estimate is surprisingly close to the exact result $Tc = 2.269185...J/k_B$. A more sophisticated version of this type of argument was first devised by Peierls to prove that in two dimensions a phase transition indeed occurs.

Probability distribution function

Definition

A random variable (also called random quantity or stochastic variable) X is a mathematical formalization of a quantity or object which depends on random events.

For a continuous random variable, the outcomes are parametrized by a continuous variable x. Then the probability for the outcome to be a value between x and $x + dx$ is expressed as

$$p(x)dx$$

normalized to

The function $p(x)$ is called the **probability distribution function** or pdf.

Moments

The expectation value of a function of x is

$$\mathbb{E}\left[f(x)\right] = \langle f(x) \rangle = \int dx\, f(x)\, p(x).$$

By definition, $\langle 1 \rangle = 1$. The m-th moment is defined as

$$\mathbb{E}\left[x^m\right] = \langle x^m \rangle = \int dx\, x^m\, p(x)$$

The mean is

$$\mathbb{E}\left[x\right] = \langle x \rangle = \int dx\, x\, p(x)$$

Moment-Generating Function

The **moment-generating function** (MGF) can be used to compute a distribution's moments. The moment generating function of a random variable X is

$$M(t) = \mathbb{E}\left[e^{tx}\right]$$

provided this expectation exists for t in some open neighborhood of 0. The moment generating function can be expanded in an infinite series in the moments of X:

$$M(t) = 1 + t\langle x\rangle + \frac{t^2}{2!}\langle x^2\rangle + \frac{t^3}{3!}\langle x^3\rangle + \cdots$$

Probability theory

The **characteristic function** is the **generator** of moments of the probability distribution. It is simply the **Fourier transform** of the PDF, defined by

The PDF can be recovered from the characteristic function through the inverse Fourier transform

Moments of the distribution are obtained by expanding $\tilde{p}(k)$ in powers of k,

The **cumulant generating function** is the **logarithm of the characteristic function**. Its expansion generates the **cumulants of the distribution** defined through

Relations between moments and cumulants can be obtained by expanding the logarithm of $\tilde{p}(k)$ using

Re-exponentiating, we finally obtain

Let us match the powers of $(-ik)^m$ on both sides:

The cumulants provide a useful and compact way of describing a PDF.

Linked cluster theorem

Linked cluster theorem

$$\langle x^m \rangle = \sum_{\{n_\ell\}}{}' \frac{m!}{\prod_\ell n_\ell!(\ell!)^{n_\ell}} \langle x^\ell \rangle_c^{n_\ell}$$

The sum is restricted such that $\sum_\ell \ell n_\ell = m$. This has a nice **graphical interpretation**: Represent the ℓ-th cumulant graphically as a connected cluster of ℓ points. The m-th moment is then obtained by summing all possible subdivisions of m points into groupings of smaller (connected or disconnected) clusters. The contribution of each subdivision to the sum is the product of the connected cumulants that it represents.

The combinatorial factor counts the number of ways of assigning m particle labels to groups of n_ℓ ℓ-clusters. A given m-particle graph can be decomposed into n_1 1-clusters, n_2 2-clusters, $\cdots$, n_ℓ ℓ-clusters, etc. It can be obtained from the total number of permutations, $m!$, after dividing by the number of equivalent assignments. Within each bin of ℓn_ℓ particles, equivalent assignments are obtained by:

- permuting the ℓ labels in each subgroup in $\ell!$ ways, for a total of $(\ell!)^{n_\ell}$ permutations;
- the $n_\ell!$ rearrangements of the n_ℓ subgroups.

The sum is restricted such that $\sum_\ell \ell n_\ell = m$.

The linked cluster theorem has the simple geometrical interpretation that the sum over all graphs, connected or not, equals the exponential of the sum over connected clusters. This theorem is the starting point for various diagrammatic computations in statistical mechanics and field theory. In the context of statistics, the first four cumulants are called the mean, variance, skewness, and curtosis (or kurtosis) of the distribution, respectively. The mean is

The variance is

The skewness is

The curtosis is

Example 6.1

Calculate cumulants of the normal (Gaussian) distribution.

The normal distribution serves as the starting point for most perturbative computations in field theory. The normal (Gaussian) distribution describes a continuous real random variable x, with

The vanishing of higher cumulants implies that all graphical computations involve only products of one-point, and two-point (known as propagators), clusters.

Wick's theorem for single variable

Consider the normal (Gaussian) distribution with $\lambda = 0$ and only $\langle x^2 \rangle_c = \sigma^2$ is non-vanishing. Gian Carlo Wick came up with a ingenious way to calculate moments using the following graphical mnemonic, now known as the **Wick's theorem**. Consider $\langle x^6 \rangle$ for normal (Gaussian) distribution with $\lambda = 0$. Write $\langle x^6 \rangle$ as $\langle xxxxxx \rangle$ and connect the x's, for example

The pattern of connection is known as a Wick contraction. In this simple example, since the six x's are identical, any one of the distinct Wick contractions gives the same value σ^6 and the final result for $\langle x^6 \rangle$ is just σ^6 times the number of distinct Wick contractions, namely $5 \cdot 3 \cdot 1 = 15$.

In this case, we see now that $\langle x^{2n+1} \rangle = 0$ since $\langle x \rangle_c = \lambda = 0$ and $\langle x^{2n} \rangle$ can be evaluated as

We will soon come to a less trivial example, with distinct x's, in which case distinct Wick contraction gives distinct values. The normal distribution serves as the starting point for most perturbative computations in field theory.

Case of many random variables

We can generalized to the case of many random variables.

The joint characteristic function is obtained from the N-dimensional Fourier transformation of the joint PDF

The joint moments and joint cumulants are generated by $\tilde{p}(k)$ and $\ln \tilde{p}(k)$, respectively, as

The previously described graphical relation between joint moments (all clusters of labeled points) and joint cumulant (connected clusters) is still applicable.

Example 6.2

Calculate cumulants of the joint Gaussian distribution.

The joint Gaussian distribution is the generalization of normal (Gaussian) distribution to N random variables, as

Wick's theorem for many variables

In the special case of $\{\lambda_m\} = 0$, all odd moments of the distribution are zero, while the general rules for relating moments to cumulants indicate that any even moment is obtained by summing over all ways of grouping the involved random variables into pairs. Recall that $\mathbf{C}$ is symmetric. As in the simple case when x does not carry any index, we could connect the x's in $\langle x_i x_j x_k x_\ell \rangle$ in pairs (Wick contraction) and simplify the calculation in terms of $\langle x_i x_j \rangle = C_{ij}$, i.e., write a factor C_{ij} if we connect x_i to x_j. Doing it is easier than explaining how to do it. You will see immediately how your result generalizes. For example, consider the case of $\langle x_a x_b x_c x_d \rangle$:

When the set of indices $i, j, \ldots, k, \ell$, contains an odd number of elements,

$$\langle x_i x_j \cdots x_k x_\ell \rangle =$$

vanishes trivially. When the set of indices $i, j, \ldots, k, \ell$ contains an even number of elements, we have

where the set of indices $\{a, b, \ldots, c, d\}$ represent a permutation of $\{i, j, \ldots, k, \ell\}$. The sum is over all such permutations or Wick contractions.

Important concepts

- Partition function is the sum of all diagrams (both connected and disconnected)
- Free energy is the sum of all connected diagrams only.
- Physical quantities are related to connected diagrams!

Low temperature expansion

Lattice models can also be studied by series expansions. Such expansions start with certain exactly solvable limits, and typically represent perturbations around such limits by graphs on the lattice. At low temperatures for models with discrete spin variables the dominant contribution to the partition function is from states where few spins are flipped relative to their value in the ground state. To exploit this we choose to order the terms in the partition function sum

where ΔZ_n is the sum of Boltzmann factors (with energy, for convenience, being measured relative to the ground state energy E_0) of all states where n spins are flipped relative to the ground state.

We describe the low-temperature expansion for the Ising model with Hamiltonian

on a 2-dimensional hypercubic lattice. The ground state with $K = \beta J > 0$ is ferromagnetic, e.g. with $S_i = +1$ for all spins. For the Ising model each wrong bond associated with flipping a spin has an energy, relative to the ground state, of $2J$ and hence a Boltzmann weight

x is the natural expansion variable for the low temperature series.

A series expansion for the partition function is obtained by including low energy excitations around this state. We can view low-energy excitations as islands of negative (positive) spins in the sea of positive (negative) spins. The zeroth order term comes from the ground state which has a two-fold degeneracy.

The lowest energy excitation is a single overturned spin. For a single spin flip, which in two dimensions generates four dissatisfied bonds, the Boltzmann weight is x^4. Any of N sites can be chosen for this excitation, which has an energy cost of $2J \times 4$, with respect to the ground state.

The second lowest energy excitation corresponds to a dimer of negative spins with energy cost of $2J \times 6$, and a multiplicity of $2N$ (there are 2 possible orientations for the dimer).

The third lowest energy excitation comes from flipping two disjoint single spins, with energy cost of $2J \times 8$ and a multiplicity of $N(N-5)/2$.

A careful analysis of the low temperature expansion yields the following:

Number of flipped spins	Configuration	Count	Boltzmann weight
1		N	x^4
2		$2N$	x^6
		$N(N-5)/2$	x^8
3		$2N$	x^8
		$4N$	x^8
		$2N(N-8)$	x^{10}
		$N(N^2-15N+62)/6$	x^{12}
4		N	x^8
		$8N$	x^{10}
		$2N$	x^{10}
		$4N$	x^{10}
		$4N$	x^{10}
		$\vdots$ (terms up to x^{16})	
5		$8N$	x^{10}
		$\vdots$ (terms up to x^{20})	
6		$2N$	x^{10}
		$\vdots$ (terms up to x^{24})	

The first few terms in the expansion give

The overall factor of 2 is insignificant in the $N \to \infty$ limit. The low T series can be written as

We can rewrite the partition function as

The free energy is obtained from the series,

- Note that there is an explicit cancellation of the term proportional to N^2 in the expansion for Z.
- Such higher N dependences result from disconnected diagrams with several disjoint droplets.
- **Extensivity** of the free energy insures that these terms are canceled by products of connected graphs. This is yet another example of the **linked cluster theorem**, used to simplify **Feynman diagram calculations**.
- Looking at the table above (Yeomans table 6.2), we see that at each order, we only need to include the part of the count that is **proportional to** N.

The energy per site is then obtained from

and the heat capacity is proportional to

Example 6.3: Low temperature expansion of the Ising model

Sethna problem 8.18

High temperature expansion

Let's begin with the high-temperature expansion of the Ising model partition function.

At high temperature, $\beta \to 0$, so we can expand

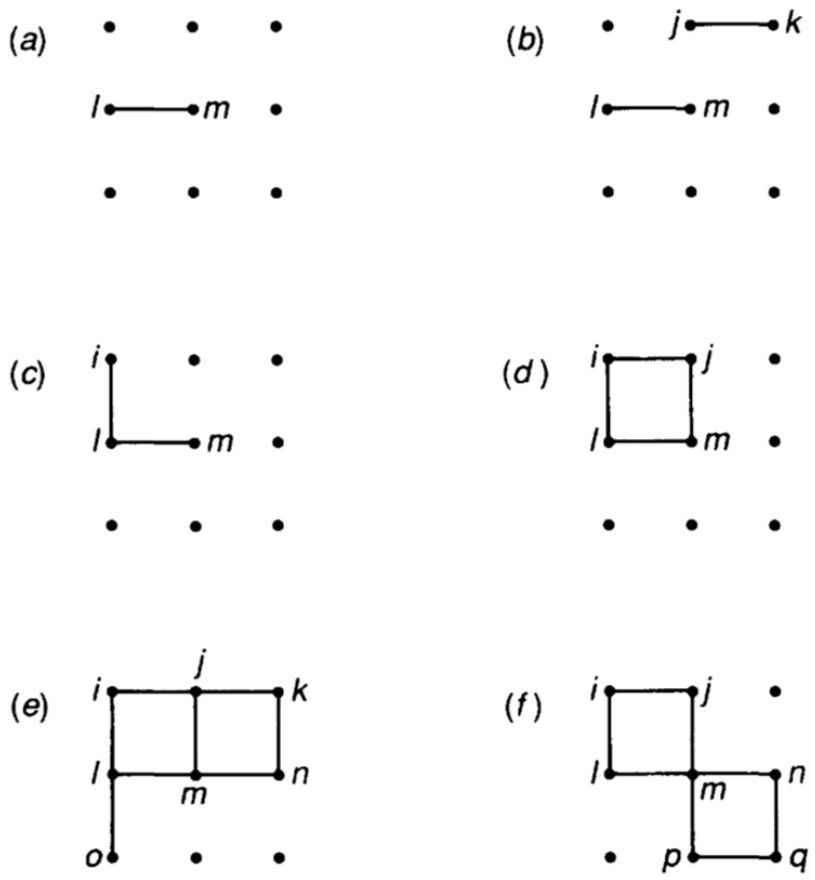
Actually, since $s_i^2 = 1$, we can simplify this expression

It is convenient to organize the high temperature expansion for the Ising model in powers of $\tanh K$. The Boltzmann factor for each bond can be written as

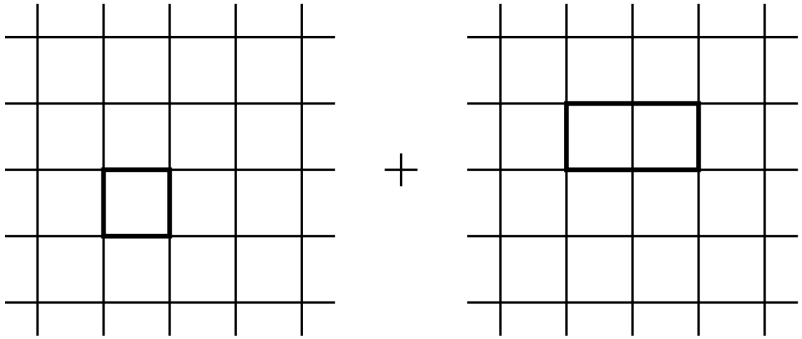
where $v \equiv \tanh K$ is a good high-temperature expansion parameter.

Applying this transformation to each bond of the lattice results in

For N_b bonds on the lattice, the above product generates 2^{N_b} terms, which can be represented diagrammatically by drawing a line connecting sites i and j for each factor of vs_is_j. Note that there can at most be one such line for each lattice bond, which is either empty or occupied.



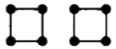
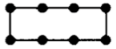
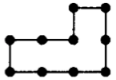
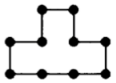
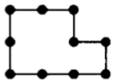
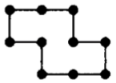
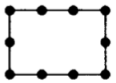
This is a major simplification, and a major advantage for the use of v, rather than K, as the expansion parameter. Each site now obtains a factor of $s_i^{p_i}$, where p_i is the number of occupied bonds emanating from i. Summing over the two possible values $s_i = \pm 1$, gives a factor of 2 if p_i is even, and 0 if p_i is odd. Thus, the only graphs that survive the sum have an even number of lines passing through each site.



For a 2-dimensional lattice, the smallest closed graph is a square of four bonds which has 2 possible orientations.

The second lowest graph has six bonds:

The third lowest graph has eight bonds:

Order	Contributing graphs	Count
v^4		N
v^6		$2N$
v^8		$N(N-5)/2$
		$4N$
		N
		$2N$
v^{10}		$2N(N-8)$
		$2N$
		$8N$
		$4N$
		$8N$
		$4N$
		$2N$

The resulting graphs are collections of closed paths on the lattice, and

The free energy per site is

The free energy is extensive as expected. Looking at the table above (Yeomans table 6.1), we see that at each order, we only need to include the part of the count that is **proportional to** N.

Example 6.4: Spin-spin correlation function

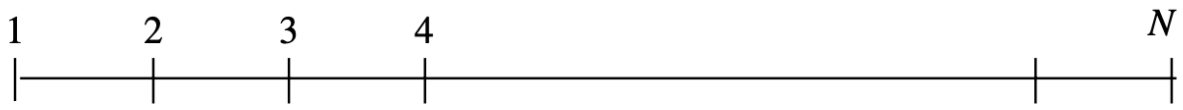
Yeomans problem 6.6

6.6 Use a high temperature series expansion to show that the two-spin correlation function of the one-dimensional Ising model in zero field is

$$\Gamma_R = \tanh^R \beta J.$$

The graphical method provides a rapid way of solving the Ising model at zero field in $d = 1$. We can compare and contrast the solutions on chains with open and closed (periodic) boundary conditions.

(1): An open chain of N sites has $N_b = N - 1$ bonds.



It is impossible to draw any closed graphs on such a lattice, and hence

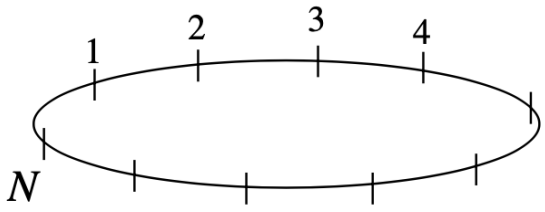
The correlation function $\langle s_m s_n \rangle$ is

The terms in the numerator involve an additional factor of $\langle s_m s_n \rangle$. To get a finite value after summing over $s_m = \pm 1$ and $s_n = \pm 1$, we have to examine graphs with an odd number of bonds emanating from these external sites. The only such graph for the open chain is one that directly connects these two sites.

This results in

The correlation length diverges as e^{2K} for $K \to \infty$, and there is no power law modifying the exponential decay of correlations.

(2): A closed chain has the same number of sites and bonds, N. It is now possible to draw a closed graph that circles the whole chain, and



The difference between the free energies of closed and open chains vanishes in the thermodynamic limit. The correction to the extensive free energy, $N \ln(2 \cosh K)$, is of the order of $1/N$ for the open chain, and can be regarded as the surface free energy. There is no such correction for the closed chain, which has no boundaries; there is instead an exponential term v^N. The correlation function can

There are two paths connecting the points m and n, and

Note that the final answer is symmetric with respect to the two ways of measuring the distance between the two sites m and n.

Table 6.3. The configurations, together with their counts and Boltzmann weights, which contribute to the low temperature expansion of the partition function of the one-dimensional Ising model

Number of flipped spins	Configuration	Count	Boltzmann weight
1	•	N	x^4
2	•—•	N	x^2
	• •	$N(N-3)/2$	x^4
3	•—•—•	N	x^2
	•—• •	$N(N-4)$	x^4
	• • •	$N(N^2-9N+20)/6$	x^6
4	•—•—•—•	N	x^2
		⋮	
		(terms up to x^8)	

Example 6.5: 2d spin 1/2 Ising model

Yeomans problem 6.7

6.7 This problem deals with the extension of the high temperature series expansion of the spin-1/2 Ising model, developed in Section 6.1, to include a field term.
(i) Show that the partition function of the spin-1/2 Ising model in a field can be written

$$\mathcal{Z} = \cosh^{\mathcal{B}} \beta J \cosh^N \beta H \sum_{\{s\}} \prod_{\langle ij \rangle} (1 + s_i s_j v) \prod_i (1 + s_i y)$$

where $v = \tanh \beta J$; $y = \tanh \beta H$, and $\mathcal{B}$ and N are the number of bonds and sites on the lattice respectively.
(ii) Show that the terms in $\mathcal{Z}$ can be represented by graphs on a lattice where each graph with l bonds and m odd vertices contributes a factor $2^N v^l y^m$, and hence that $\mathcal{Z}$ can be rewritten

$$\mathcal{Z} = \cosh^{\mathcal{B}} \beta J (2 \cosh \beta H)^N (1 + \mathcal{S}_0 + y^2 \mathcal{S}_2 + y^4 \mathcal{S}_4 + \ldots) \quad (6.21)$$

where $\mathcal{S}_m(v, N)$ is the contribution from all graphs with m odd vertices.
Taking the logarithm of eqn (6.21) and using the Linked Cluster Theorem[6] gives

$$-\beta \mathcal{F} = \mathcal{B} \ln \cosh \beta J + N \ln 2 \cosh \beta H + (\mathcal{S}_0' + y^2 \mathcal{S}_2' + y^4 \mathcal{S}_4' + \ldots) \quad (6.22)$$

where the prime denotes that, for a given graph, only the part of the count proportional to N must be included.
(iii) Differentiate eqn (6.22) to show that the zero-field susceptibility is

$$\chi = \beta N + 2\beta \mathcal{S}_2'.$$

(iv) List the low order graphs with two odd vertices to show that the first few terms in the high temperature expansion of the zero-field susceptibility of the spin-1/2 Ising model on the square lattice are

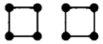
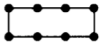
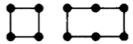
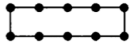
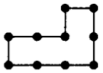
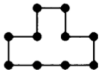
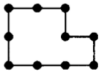
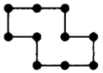
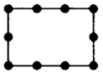
$$\chi = \beta N + 2\beta (2v + 6v^2 + 18v^3 + \ldots).$$

Self-duality in the two-dimensional Ising model

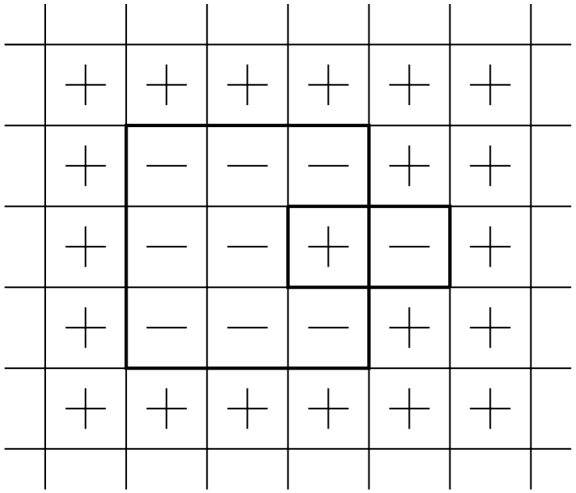
Kramers and Wannier discovered a hidden symmetry that relates the properties of the Ising model on the square lattice at low and high temperatures. One way of obtaining this symmetry is to compare the high and low temperature series expansions of the problem. The low-temperature expansion has the form

The high-temperature series is

Number of flipped spins	Configuration	Count	Boltzmann weight
1		N	x^4
2		$2N$	x^6
		$N(N-5)/2$	x^8
3		$2N$	x^8
		$4N$	x^8
		$2N(N-8)$	x^{10}
		$N(N^2-15N+62)/6$	x^{12}
4		N	x^8
		$8N$	x^{10}
		$2N$	x^{10}
		$4N$	x^{10}
		$4N$	x^{10}
	$\vdots$ (terms up to x^{16})		
5		$8N$	x^{10}
	$\vdots$ (terms up to x^{20})		
6		$2N$	x^{10}
	$\vdots$ (terms up to x^{24})		

Order	Contributing graphs	Count
v^4		N
v^6		$2N$
v^8		$N(N-5)/2$
		$4N$
		N
		$2N$
v^{10}		$2N(N-8)$
		$2N$
		$8N$
		$4N$
		$8N$
		$4N$
		$2N$

As the boundary of any island of spins serves as an acceptable graph (and vice versa), there is a one-to-one correspondence between the two series.

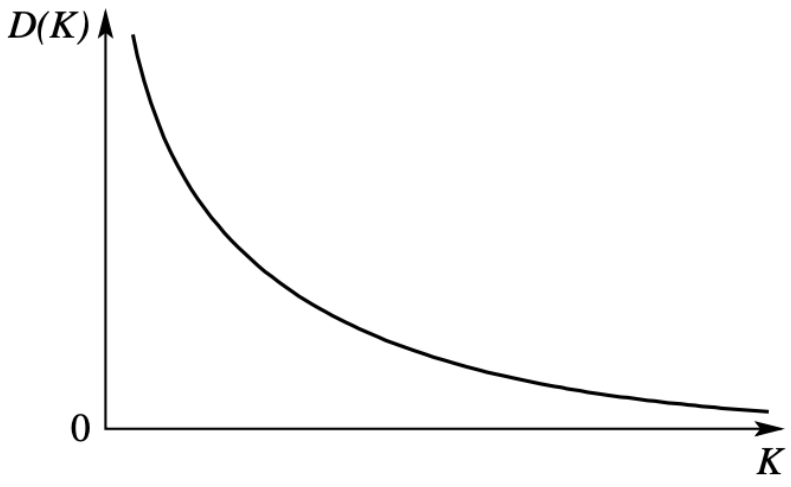


Defining a function g to indicate the logarithm of the above series, the free energy is given by

The arguments of g in the above equation are related by the duality condition

The function g (which contains the singular part of the free energy) must have a special symmetry that relates its values for dual arguments. (For example the function $f(x) = x/(1+x^2)$ equals $f(x^{-1})$, establishing a duality between the arguments at x and x^{-1}.)

The function $D(K)$ relating the parameter K to its dual value:



- Low temperatures are mapped to high temperatures, and vice versa.
- The mapping connects pairs of points since $D(D(K)) = K$. This condition is established by using trigonometric identities to show that

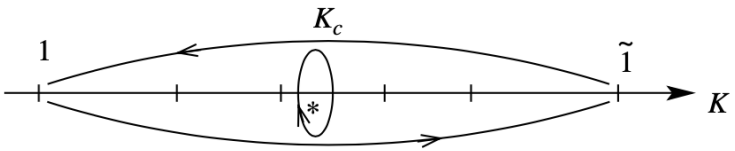
Hence, the dual interactions are symmetrically related by

- If the function $g(K)$ is singular at a point K_c, it must also be singular at $\tilde{K}_c$. Since the free energy is expected to be analytic everywhere except at the transition, the critical model must be self-dual. At the self-dual point

  which leads to the quadratic equation,

  Only the positive solution is acceptable, and

The duality function maps pairs of points from high and low temperatures. The self-dual point at K_c maps onto itself.



- In 1944, in a tour de force calculation, **Lars Onsager** (Nobel Prize in Chemistry 1968) exactly diagonalized the transfer matrix for 2d Ising model. He showed that the free energy $F(T)$ has one singularity and it occurs at the critical point $k_B T_c = 2J/\log(\sqrt{2}+1)$, corresponding to the Kramers-Wannier self-dual point! This exact solution is described in *TD Schultz, DC Mattis, EH Lieb, Reviews of Modern Physics,* **36**, *856 (1964).*
- If we apply same procedure to search for the dual of the Ising model on a simple cubic lattice, we will find that the 3d Ising model is not dual to itself.

Analysis of series expansion

Suppose that we have identified a number of terms in a series

From the expected divergence of the heat capacity, we expect an asymptotic expansion of the form

The above singular form is characterized by the three parameters A, u_c, and α.

We can try to extract these parameters from the calculated coefficients in the series by requiring that they match at large ℓ, i.e. by fitting the ratio of successive terms to

Thus a plot of $a_\ell/a_{\ell-1}$ versus $1/\ell$ should be a straight line with intercept u_c^{-1}, and slope $u_c^{-1}(\alpha-1)$. Note, however, that adding a finite sum

does not change the asymptotic singular form, but essentially renders the first ℓ_m terms useless in determining α and u_c^{-1}. There is thus no a priori guarantee that such a fitting procedure will succeed with a finite number of coefficients. In practice, this procedure works very well and very good estimates of critical exponents in $d=3$ (e.g. $\alpha = 0.105 \pm 0.007$) are obtained in this way by including a rather large number of terms.

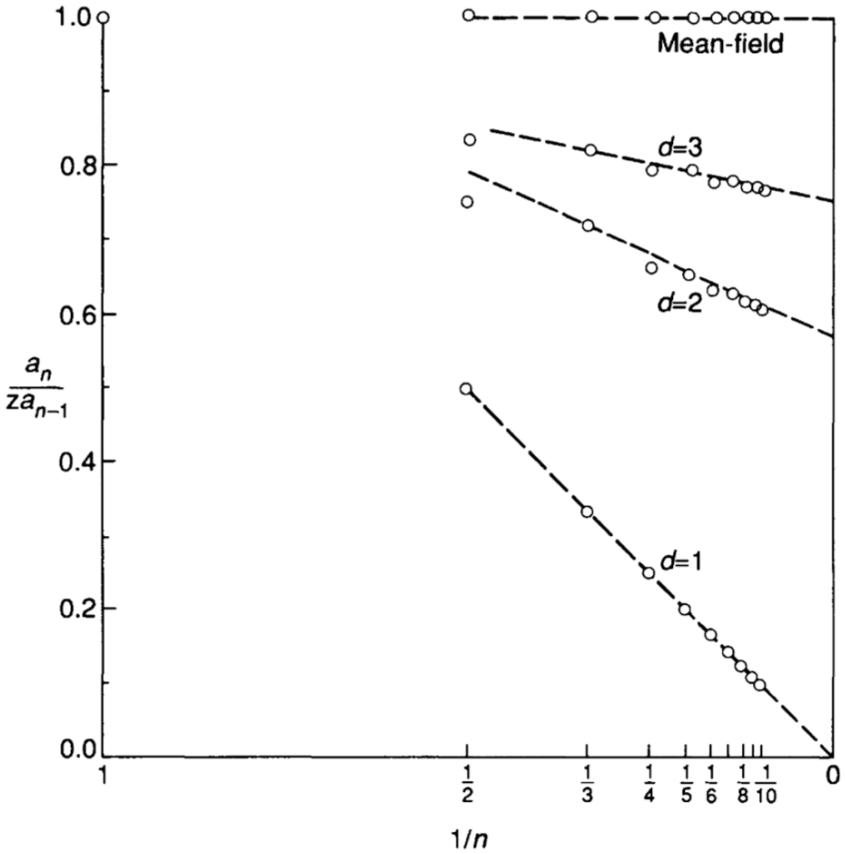


Figure: Fig 6.2 of Yeomans: plot of a_n/za_{n-1} of the reduced susceptibility series of spin 1/2 Ising model on lattices of different dimensionality plotted against $1/n$.

Extra notes